

FitBuddy – AI Fitness Plan Generator

Project Design Phase – Page 1

Introduction

The Design Phase converts the requirements of FitBuddy into a clear system structure. The design describes how the frontend, backend, AI services, database and user interactions work together.

Design Objectives

The main objective is to create a simple, responsive and understandable fitness-planning application. The design should support personalized workout generation, nutrition guidance, feedback-based plan updating and an administrative view.

Main Design Areas

- System architecture and module structure
- User interface and page design
- Backend routes and processing flow
- Gemini AI integration
- Database and data storage
- Feedback and plan-update mechanism
- Deployment and configuration design

FitBuddy – AI Fitness Plan Generator

Project Design Phase – Page 2

Architecture

FitBuddy is designed as a web application with separate presentation, application, AI-service and database responsibilities. The user interacts with the frontend, while the backend processes requests and communicates with Gemini and the database.

Major Components

- Frontend – HTML/Jinja2 templates for user interaction and displaying results
- FastAPI backend – handles application routes and request processing
- Gemini AI layer – generates workout plans and nutrition tips
- Database layer – stores users and generated plans using SQLite/SQLAlchemy
- Configuration layer – manages the Gemini API key and application settings

FitBuddy – AI Fitness Plan Generator

Project Design Phase – Page 3

Workout Generation Module

The workout module accepts user information such as age, weight, fitness goal and workout intensity. It sends the required information to the Gemini-based generator and receives a personalized 7-day workout plan.

Nutrition Module

The nutrition component generates a suitable nutrition or recovery tip based on the user's fitness context.

Feedback Module

Users can submit feedback such as requesting more cardio or adding yoga. The feedback is processed to create an updated version of the original plan.

User and Admin Module

User information and plans are stored for later access. The administrative view displays users and their original or updated plans.

FitBuddy – AI Fitness Plan Generator

Project Design Phase – Page 4

Home/Input Page

The main page is designed to collect the information required for personalized plan generation. It includes fields for user details, fitness goal and workout intensity.

Result Page

The result page displays the generated workout plan and nutrition tip in a clear format. It also provides a way for the user to submit feedback for modifications.

Admin/User View

The all-users page is designed to display stored user information and their generated plans in an organized manner.

Design Principles

- Simple navigation
- Clear labels and input fields
- Readable workout results
- Responsive and clean layout
- Easy feedback submission
- Consistent page structure

FitBuddy – AI Fitness Plan Generator

Project Design Phase – Page 5

FastAPI Backend

The backend acts as the central controller of the application. It receives user requests, validates and processes input, calls the required functions, stores data and returns the appropriate page.

Main Routes

- / – displays the main input page
- /generate-workout – receives user details and generates the workout plan
- /submit-feedback – receives feedback and updates the plan
- /view-all-users – displays stored user and plan information

Processing Sequence

A typical request follows this sequence: user input → FastAPI route → validation/processing → AI generation → database storage → result page.

FitBuddy – AI Fitness Plan Generator

Project Design Phase – Page 6

Gemini Workout Generation

The design uses a Gemini model to generate a structured 7-day workout plan according to the user's fitness goal and workout intensity. The generator function is separated from the route logic so that the AI functionality can be maintained independently.

Nutrition Tip Generation

A separate Gemini-based function is used for quick nutrition or recovery guidance. This keeps the nutrition generation responsibility distinct from workout generation.

Feedback-Based Updating

The update module receives the original plan and the user's feedback and generates a modified plan. The updated plan is then stored and displayed to the user.

AI Design Considerations

- Use structured prompts
- Pass only required user information
- Keep AI functions separate from routing
- Handle API failures appropriately
- Store generated results for later viewing

FitBuddy – AI Fitness Plan Generator

Project Design Phase – Page 7

Database Purpose

The database design provides persistent storage for user details and generated workout plans. SQLite is suitable for the project while SQLAlchemy provides database interaction.

Main Data

- User identification information
- Age and weight
- Fitness goal
- Workout intensity
- Original generated workout plan
- Updated workout plan
- Feedback used for plan revision

Database Functions

The project design includes functions for saving users, saving plans, updating plans and retrieving users or original plans. Separating these functions keeps database operations organized.

Data Flow

User data is submitted → user record is saved → generated plan is saved → feedback may update the plan → updated result can be retrieved and displayed.

FitBuddy – AI Fitness Plan Generator

Project Design Phase – Page 8

Overall Workflow

The system begins when a user enters personal fitness information. The application sends the information through the backend to the AI generation layer. The generated plan is stored and displayed. If feedback is provided, the plan is revised and the new version is stored.

Step-by-Step Flow

- 1. User opens FitBuddy
- 2. User enters required fitness details
- 3. Backend receives the request
- 4. Workout generator creates a 7-day plan
- 5. Nutrition tip is generated
- 6. User and plan information are stored
- 7. Result is displayed
- 8. User may submit feedback
- 9. Updated plan is generated and stored
- 10. Updated plan is displayed

FitBuddy – AI Fitness Plan Generator

Project Design Phase – Page 9

Configuration and Security

The Gemini API key is designed to be stored through environment configuration rather than directly exposing it in application pages or source code. The application should also validate submitted form data before processing it.

Error Handling

- Handle missing or invalid user input
- Handle unavailable AI/API responses
- Show understandable error messages
- Prevent application crashes where possible
- Keep database operations controlled

Deployment Design

The project is designed to run in a Python virtual environment with the required dependencies installed. The FastAPI application can be served using Uvicorn. Environment variables are used for configuration such as the Gemini API key.

FitBuddy – AI Fitness Plan Generator

Project Design Phase – Page 10

Integrated Design

The FitBuddy design connects the user interface, FastAPI backend, Gemini AI services and database into one workflow. Each major responsibility is separated into modules so the application can be developed, tested and maintained more easily.

Expected System Behaviour

- Accept user fitness information
- Generate a personalized 7-day workout plan
- Provide a nutrition/recovery tip
- Store user and plan information
- Accept feedback and update the plan
- Display original and updated plans
- Provide an organized user/admin view

Conclusion

The Design Phase provides the technical blueprint for implementing FitBuddy. It defines the architecture, modules, interfaces, routes, AI integration, database structure, workflow and deployment approach. This design can be used as the foundation for the Implementation Phase.